

$$1) \begin{aligned} a_7 &= 20 \\ a_{31} &= 92 \\ \hline a_1, d, s_{10} &=? \end{aligned}$$

$$a_{31} = a_7 + (31-7)d \quad a_7 = a_1 + (7-1)d$$

$$a_{31} = a_7 + 24d$$

$$92 = 20 + 24d$$

$$72 = 24d$$

$$\underline{\underline{3 = d}}$$

$$a_7 = a_1 + 6d$$

$$20 = a_1 + 6 \cdot 3$$

$$20 = a_1 + 18$$

$$\underline{\underline{2 = a_1}}$$

$$s_{10} = \frac{10}{2} (2 + 29)$$

$$s_{10} = 5 \cdot 31$$

$$\underline{\underline{s_{10} = 155}}$$

$$a_8 = a_7 + d$$

+1

→ rozdíl drach  
členů

$$a_{10} = a_1 + 9d$$

$$a_{10} = 2 + 27$$

$$\underline{\underline{a_{10} = 29}}$$

$$2) \begin{aligned} a_1 + a_5 &= -8 \\ a_2 + a_6 &= -4 \end{aligned}$$

$$\underline{\underline{a_1; d = ?}}$$

$$a_5 = a_1 + 4d$$

$$a_6 = a_1 + 5d$$

$$a_2 = a_1 + d$$

$$a_n = a_1 + (n-1)d$$

$$a_1 + a_1 + 4d = -8$$

$$a_1 + d + a_1 + 5d = -4$$

$$\underline{2a_1 + 4d = -8}$$

$$\underline{2a_1 + 6d = -4} \quad / \cdot (-1)$$

$$-2a_1 - 6d = 4$$

$$\underline{-2d = -4}$$

$$\underline{\underline{d = 2}}$$

$$2a_1 + 4 \cdot 2 = -8$$

$$2a_1 + 8 = -8$$

$$2a_1 = -16$$

$$\underline{\underline{a_1 = -8}}$$